More complex mixed models:

Repeated records
Common Environment
Maternal Effect

Assessment of random effects

Repeatability model

- Multiple measurements on the same trait (phenotype) are recorded on an individual; for example:
  - Height at different ages
  - Production (milk yield) in successive time periods (cows)
- Variation in and similarities between repeated records on the same individual is attributed to both genetic and environmental factors consistently affecting the successive phenotypic measurements (permanent environmental effects).
- Such environmental effects should be accounted for in the model to ensure accurate prediction of the genetic effect.

Basic QGM model

Phenotype = Genotype + "Environment"

$$P = G + E$$

$$P = A+D+I + E_{perm} + E_{temp}$$

Permanent environmental effect on P
e.g. a chronic condition

Temporary environmental effect on P
e.g. weather event, one-off treatment

Repeatability model

- **Some assumptions:**
  - **Genetic correlations between successive records (phenotypic measurements on the same individual) = 1**
  - **Equal variance for all records**
  - **Equal environmental correlations among records**
- ***Assumptions are relaxed with the use of random regression models for the analysis of such data***

Previous models

- **The phenotypic structure for 3 observations of an individual:**

$$var\begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \mathbf{y}_3 \end{bmatrix} = \begin{bmatrix} \sigma_g^2 + \sigma_{e1}^2 & \sigma_g^2 & \sigma_g^2 \\ \sigma_g^2 & \sigma_g^2 + \sigma_{e2}^2 & \sigma_g^2 \\ \sigma_g^2 & \sigma_g^2 & \sigma_g^2 + \sigma_{e3}^2 \end{bmatrix}$$

Variation in each record determined by variation in genotype and residual

$\sigma^2_g = \sigma^2_u$ same thing (genetic variance), different notation

$\sigma^2_p = \sigma^2_y$ same thing (phenotypic variance), different notation

Previous models

- **The phenotypic structure for 3 observations of an individual:**

$$var\begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \mathbf{y}_3 \end{bmatrix} = \begin{bmatrix} \boldsymbol{\sigma}_{\mathbf{g}}^2 + \boldsymbol{\sigma}_{e1}^2 & \boldsymbol{\sigma}_{\mathbf{g}}^2 & \boldsymbol{\sigma}_{\mathbf{g}}^2 \\ \boldsymbol{\sigma}_{\mathbf{g}}^2 & \boldsymbol{\sigma}_{\mathbf{g}}^2 + \boldsymbol{\sigma}_{e2}^2 & \boldsymbol{\sigma}_{\mathbf{g}}^2 \\ \boldsymbol{\sigma}_{\mathbf{g}}^2 & \boldsymbol{\sigma}_{\mathbf{g}}^2 & \boldsymbol{\sigma}_{\mathbf{g}}^2 + \boldsymbol{\sigma}_{e3}^2 \end{bmatrix}$$

Covariances (similarities) between records attributed to the genetic effect

$\sigma^2_g = \sigma^2_u$ same thing (genetic variance), different notation

$\sigma^2_p = \sigma^2_y$ same thing (phenotypic variance), different notation

Repeatability model

- The phenotypic structure for 3 observations of an individual under the repeatability model is:

$$var\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} \sigma_g^2 + \sigma_{pe}^2 + \sigma_{e1}^2 & \sigma_{pe}^2 + \sigma_g^2 & \sigma_{pe}^2 + \sigma_g^2 \\ \sigma_{pe}^2 + \sigma_g^2 & \sigma_g^2 + \sigma_{pe}^2 + \sigma_{e2}^2 & \sigma_{pe}^2 + \sigma_g^2 \\ \sigma_{pe}^2 + \sigma_g^2 & \sigma_{pe}^2 + \sigma_g^2 & \sigma_g^2 + \sigma_{pe}^2 + \sigma_{e3}^2 \end{bmatrix}$$

Covariance (similarities) between records attributed to the genetic and permanent environmental effects

- Heritability $h^2 = \sigma^2_g / \sigma^2_{y/p}$

- Repeatability $r_e = (\sigma^2_g + \sigma^2_{pe}) / \sigma^2_{y/p}$

Repeatability model

The model then is:

$$y = Xb + Zu + Wpe + e$$

y = vector of observations
b = vector of fixed effects
u = vector of random effects (individual/genetic)
pe = vector of random PE effects
e = vector of random residual effect
X, Z, W = incidence matrices

Repeatability model

- **The MME are:**

$$\begin{bmatrix} \mathbf{X'X} & \mathbf{X'Z} & \mathbf{X'W} \\ \mathbf{Z'X} & \mathbf{Z'Z}+\mathbf{A}^{-1}\lambda_1 & \mathbf{Z'W} \\ \mathbf{W'X} & W'\mathbf{Z} & \mathbf{W'W}+\mathbf{I}\lambda_2 \end{bmatrix} \begin{bmatrix} \hat{\mathbf{b}} \\ \hat{\mathbf{u}} \\ \mathbf{p}\hat{\mathbf{e}} \end{bmatrix} = \begin{bmatrix} \mathbf{X'Y} \\ \mathbf{Z'Y} \\ \mathbf{W'Y} \end{bmatrix}$$

$$\lambda_1 = \frac{\sigma_e^2}{\sigma_u^2} = \frac{1-r_e}{h^2} \qquad \lambda_2 = \frac{\sigma_e^2}{\sigma_{pe}^2} = \frac{\sigma_e^2}{\left(r_e - h^2\right)\sigma_P^2} = \frac{1-r_e}{r_e - h^2}$$

Repeatability Model (example)

Cow	Sire	Dam	Parity	Cohort	Milkfat yield (kg)
4	1	2	1	1	201
4	1	2	2	3	280
5	3	2	1	1	150
5	3	2	2	4	200
6	1	5	1	2	160
6	1	5	2	3	190
7	3	4	1	1	180
7	3	4	2	3	250
8	1	7	1	2	285
8	1	7	2	4	300

y
10 x 1

Assume/ estimate:

σ^2_u=20, σ^2_e= 28 and σ^2_{pe}= 12 (total phenotypic σ^2_y= 60)

h^2 = 0.33 r_e = 0.53

$\lambda_1 = \sigma^2_e/\sigma^2_u$= 28/20 = 1.40

$\lambda_2 = \sigma^2_e/\sigma^2_{pe}$= 28/12 = 2.33

Fixed effects

Cohort: 4 levels
Parity: 2 levels

$$X' = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ \hline 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

6×10

$$X'X = \begin{bmatrix} 3 & 0 & 0 & 0 & 3 & 0 \\ 0 & 2 & 0 & 0 & 2 & 0 \\ 0 & 0 & 3 & 0 & 0 & 3 \\ 0 & 0 & 0 & 2 & 0 & 2 \\ 3 & 2 & 0 & 0 & 5 & 0 \\ 0 & 0 & 3 & 2 & 0 & 5 \end{bmatrix}$$

6×6

$$X'y = \begin{bmatrix} 531 \\ 445 \\ 720 \\ 500 \\ \hline 976 \\ 1220 \end{bmatrix}$$

6×1

Random effects - genetic

Individuals:
w/records: 5
w/o records: 3

$$Z'Z = diag[0\ \ 0\ \ 0\ \ 2\ \ 2\ \ 2\ \ 2\ \ 2]$$

$$8 \times 8$$

$$Z' = \begin{bmatrix} 0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0 \\ \hline 1&1&0&0&0&0&0&0&0&0 \\ 0&0&1&1&0&0&0&0&0&0 \\ 0&0&0&0&1&1&0&0&0&0 \\ 0&0&0&0&0&0&1&1&0&0 \\ 0&0&0&0&0&0&0&0&1&1 \end{bmatrix}$$

$$8 \times 10$$

$$A^{-1}\lambda_1 = \begin{bmatrix} 2.50 & 0.50 & 0.00 & -1.00 & 0.50 & -1.00 & 0.50 & -1.00 \\ 0.50 & 1.50 & 0.00 & -1.00 & 0.00 & 0.00 & 0.00 & 0.00 \\ 0.00 & 0.00 & 1.83 & 0.50 & -0.67 & 0.00 & -1.00 & 0.00 \\ -1.00 & -1.00 & 0.50 & 2.50 & 0.00 & 0.00 & -1.00 & 0.00 \\ 0.50 & 0.00 & -0.67 & 0.00 & 1.83 & -1.00 & 0.00 & 0.00 \\ -1.00 & 0.00 & 0.00 & 0.00 & -1.00 & 2.00 & 0.00 & 0.00 \\ 0.50 & 0.00 & -1.00 & -1.00 & 0.00 & 0.00 & 2.50 & -1.00 \\ -1.00 & 0.00 & 0.0 & 0.00 & 0.00 & 0.00 & -1.00 & 2.00 \end{bmatrix} * [1.40]$$

$$8 \times 8$$

$$\text{transpose of } Z'y = [0\ \ 0\ \ 0\ \ 481\ \ 350\ \ 350\ \ 430\ \ 585]$$

$$1 \times 8$$

Random effects - PE

Individuals w/records: 5

$$W'W = diag\begin{bmatrix}2 & 2 & 2 & 2 & 2\end{bmatrix}$$

$$5 \times 5$$

$$W' = \begin{bmatrix} 1&1&0&0&0&0&0&0&0&0 \\ 0&0&1&1&0&0&0&0&0&0 \\ 0&0&0&0&1&1&0&0&0&0 \\ 0&0&0&0&0&0&1&1&0&0 \\ 0&0&0&0&0&0&0&0&1&1 \end{bmatrix}$$

$$5 \times 10$$

$$I\lambda_2 = \begin{bmatrix} 1&0&0&0&0 \\ 0&1&0&0&0 \\ 0&0&1&0&0 \\ 0&0&0&1&0 \\ 0&0&0&0&1 \end{bmatrix} * [2.33]$$

$$5 \times 5$$

$$transpose\ of\ W'y = \begin{bmatrix}481 & 350 & 350 & 430 & 585\end{bmatrix}$$

$$1 \times 5$$

Solutions to MME

$$\begin{bmatrix} \mathbf{X'X} & \mathbf{X'Z} & \mathbf{X'W} \\ \mathbf{Z'X} & \mathbf{Z'Z}+\mathbf{A}^{-1}\lambda_1 & \mathbf{Z'W} \\ \mathbf{W'X} & W'\mathbf{Z} & \mathbf{W'W}+\mathbf{I}\lambda_2 \end{bmatrix} \begin{bmatrix} \hat{\mathbf{b}} \\ \hat{\mathbf{u}} \\ \mathbf{p}\hat{\mathbf{e}} \end{bmatrix} = \begin{bmatrix} \mathbf{X'Y} \\ \mathbf{Z'Y} \\ \mathbf{W'Y} \end{bmatrix}$$

Cohort

b_1	0.00
b_2	44.07
b_3	0.00
b_4	0.01

Parity

b_5	0.00
b_6	66.42

Genetic

u_1	10.15
u_2	-3.08
u_3	-7.06
u_4	13.58
u_5	-18.21
u_6	-18.39
u_7	9.33
u_8	24.19

PE

pe_4	8.42
pe_5	-7.15
pe_6	-17.23
pe_7	-1.39
pe_8	17.35

Models with common environment

- When members of a family are reared together, they share a common environment
- This contributes to the similarity between individual performance (phenotypes)
- Additionally to genetics, there is also a covariance between members of a family due to the common environment they share
- Environmental variance may then be partitioned into:
  - between family or group component (σ^2_c), usually termed the common environment which causes resemblance between members of a family
  - and the within family or group variance (σ^2_e) = residual

Model with common environment

- Phenotypic variance (σ^2_y or σ^2_p) would then consist of:
  - variance of additive genetic effects resulting from additive genes from parents (σ^2_u)
  - variance of common environmental effects affecting all offspring raised by the same mother (σ^2_c)
  - variance of residual effects (σ^2_e)

Model with common environment

Conceptually different from permanent environment:

- **Permanent environment:** one individual; similarities between successive phenotypic records of same individual (repeatability model)
- **Common environment:** multiple individuals; similarities between contemporaneous records of multiple individuals sharing the common environment

Model with common environment

In matrix notation, the model which is

$$y = Xb + Zu + Wc + e$$

- where all terms are as defined for the repeatability model except c which is the vector of common environmental effects
- W now relates records to common environmental effects.
- It is assumed that common environmental and residual effects are independently distributed with means of zero and variance σ^2_c and σ^2_e respectively
- Thus var(c) = $I\sigma^2_c$, var(e) = $I\sigma^2_e$ and var(u) = $A\sigma^2_u$

Mixed model equations

- **The MME are:**

$$\begin{bmatrix} \mathbf{X'X} & \mathbf{X'Z} & \mathbf{X'W} \\ \mathbf{Z'X} & \mathbf{Z'Z}+\mathbf{A}^{-1}\lambda_1 & \mathbf{Z'W} \\ \mathbf{W'X} & W'\mathbf{Z} & \mathbf{W'W}+\mathbf{I}\lambda_2 \end{bmatrix} \begin{bmatrix} \hat{\mathbf{b}} \\ \hat{\mathbf{u}} \\ \hat{\mathbf{c}} \end{bmatrix} = \begin{bmatrix} \mathbf{X'Y} \\ \mathbf{Z'Y} \\ \mathbf{W'Y} \end{bmatrix}$$

$$\lambda_1 = \frac{\sigma_e^2}{\sigma_u^2} \qquad \lambda_2 = \frac{\sigma_e^2}{\sigma_c^2}$$

Example data

Piglet	sire	dam	family	gender	150-d Weight (kg)
6	1	2	1	M	90
7	1	2	1	F	70
8	1	2	1	F	65
9	3	4	2	F	98
10	3	4	2	M	106
11	3	4	2	F	60
12	3	4	2	F	80
13	1	5	3	M	100
14	1	5	3	F	85
15	1	5	3	M	68

Example data

$$Z = diag\left[0\ 0\ 0\ 0\ 0\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\right]$$

$$X' = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 0 \end{bmatrix}$$

$$W' = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

Gender		Genetic				Common env.		dam
male	91.49	u_1	-1.44	u_6	-1.10	c_1	-1.76 →	2
female	75.76	u_2	-1.18	u_7	-1.67	c_2	2.16 →	4
		u_3	1.44	u_8	-2.33	c_3	-0.40 →	5
		u_4	1.44	u_9	3.93			
		u_5	-0.27	u_{10}	2.90			
				u_{11}	-1.14			
				u_{12}	1.53			
				u_{13}	0.45			
				u_{14}	0.55			
				u_{15}	-3.82			

Maternal effect models

- The phenotypic expression of some traits in the progeny is influenced by the ability of the mother (dam) to provide a suitable environment such as better nourishment.
- Thus the mother contributes to the performance of the progeny in two ways:
  - The direct genetic effects passed onto the progeny
  - Her ability to provide a suitable environment (nurturing, nourishing)
- Examples: Body weight of offspring at birth and in early life

Maternal effect models

- The ability of the mother (dam) to provide a suitable environment for the expression of such traits in her progeny is partly genetic and partly environmental.

- The environmental part may be partitioned into <u>common (maternal)</u> and <u>residual</u> components (previous model).

- Maternal additive genetic component of the dam is passed onto her offspring but is expressed only when the female offspring have progeny of their own.

Maternal effect models

- Phenotypic variance is partitioned into:
  - Additive genetic variance – direct effect
  - Additive genetic variance of the ability of the mother (dam) to nurture – indirect or maternal genetic effect
  - Common (maternal) environment
  - Residual effects

Maternal effects model

$$y = Xb + Zu + Wm + Sc + e$$

where

- y = vector of observations
- b = vector of fixed effects
- u = vector of random direct genetic effects
- m = vector of random indirect genetic (maternal) effects
- c = vector of common environmental
- e = vector of random residual effects
- X, Z, W and S are incidence matrices

Maternal effects model

$$var\begin{bmatrix} u \\ m \\ c \\ e \end{bmatrix} = \begin{bmatrix} g_{11}A & g_{12}A & 0 & 0 \\ g_{21}A & g_{22}A & 0 & 0 \\ 0 & 0 & I\sigma_c^2 & 0 \\ 0 & 0 & 0 & I\sigma_e^2 \end{bmatrix}$$

g_{11} = additive genetic variance (direct effects)
g_{22} = additive genetic variance (maternal effects)
$g_{12} = g_{21}$ = additive genetic covariance between direct and maternal effects

Maternal effects model: MME

$$\begin{bmatrix} \hat{b} \\ \hat{u} \\ \hat{m} \\ \hat{c} \end{bmatrix} = \begin{bmatrix} X'X & X'Z & X'W & X'S \\ Z'X & Z'Z + A^{-1}\lambda_1 & Z'W + A^{-1}\lambda_2 & Z'S \\ W'X & W'Z + A^{-1}\lambda_2 & W'W + A^{-1}\lambda_3 & W'S \\ S'X & S'Z & S'W & S'S + I\lambda_4 \end{bmatrix}^{-1} \begin{bmatrix} X'y \\ Z'y \\ W'y \\ S'y \end{bmatrix}$$

$$G = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix}; G^{-1} = \begin{bmatrix} g^{11} & g^{12} \\ g^{21} & g^{22} \end{bmatrix} \text{ and } \begin{bmatrix} \lambda_1 & \lambda_2 \\ \lambda_2 & \lambda_3 \end{bmatrix} = \sigma_e^2 \begin{bmatrix} g^{11} & g^{12} \\ g^{21} & g^{22} \end{bmatrix}$$

Maternal effects model: Example

Calf	Sire	Dam	Herd	Sex	Birth weight (kg)
5	1	2	1	F	35.0
6	3	2	1	F	20.0
7	4	6	1	F	25.0
8	3	5	1	M	40.0
9	1	6	2	M	42.0
10	3	2	2	F	22.0
11	3	7	2	F	35.0
12	8	7	3	F	34.0
13	9	2	3	M	20.0
14	3	6	3	F	40.0

Maternal effects model: Example

$$X' = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ \hline 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 0 & 1 \end{bmatrix}$$

$$\mathbf{W} = \begin{array}{c} \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \\ 10 \\ 11 \\ 12 \\ 13 \\ 14 \end{array} \begin{array}{c} \begin{array}{cccccccccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

$$\mathbf{S}' = \begin{array}{c} \\ 2 \\ 5 \\ 6 \\ 7 \end{array} \begin{array}{c} \begin{array}{cccccccccc} 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix} \end{array}$$

Maternal effects model: Example

Effects	Solutions (BLUE)
Herd	
1	0.000
2	3.386
3	1.434
Sex of calf	
M	0.000
F	-6.909

Maternal effects model: Example

Individual	Direct	Maternal
1	0.564	0.262
2	-1.244	-1.583
3	1.165	0.736
4	-0.484	0.586
5	0.630	-0.507
6	-0.859	0.841
7	-1.156	1.299
8	1.917	-0.158
9	-0.553	0.660
10	-1.055	-0.153
11	0.385	0.916
12	0.863	0.442
13	-2.980	0.093
14	1.751	0.362

Common environment

Dam	
2	-1.701
5	0.415
6	0.825
7	0.461

Maternal effects model: Example

Individual	Direct	Maternal
1	0.564	0.262
2	-1.244	-1.583
3	1.165	0.736
4	-0.484	0.586
5	0.630	-0.507
6	-0.859	0.841
7	-1.156	1.299
8	1.917	-0.158
9	-0.553	0.660
10	-1.055	-0.153
11	0.385	0.916
12	0.863	0.442
13	-2.980	0.093
14	1.751	0.362

Common environment

Dam	
2	-1.701
5	0.415
6	0.825
7	0.461

Assessment of random effects

- Likelihood: A function of the parameters of the model
- ASReml yields estimates of this function for each model
- Model 1
  - number of parameters estimated: r_1
  - Log-likelihood: L_{r1}
- Model 2 (same fixed effects)
  - number of parameters estimated: r_2
  - Log-likelihood: L_{r2}
- Log-likelihood test
  - $D = 2\log(L_{r2}/L_{r1}) = 2[\log(L_{r2})-\log(L_{r1})]$
  - Chi-square with d.f. = $r_2 - r_1$
  - P-value = 0.05 means essentially the test is done at 0.025 level because of boundary problems (i.e. no negative values in variance estimates are allowed so they are forced to be zero)
- More in tutorial

Restricted Maximum likelihood (REML)

- BLUP assumes that the genetic and phenotypic parameters are known (without error).
- Therefore, the required genetic parameters need to be estimated.
- Genetic parameters are based on estimates of variance components
  - Partitioning the total phenotypic variance into components for each random effect in the model
- Variance component estimation is based on the principles of Maximum Likelihood (ML)
  - i.e. estimates that maximise the Likelihood
- When fixed effects are accounted for in a mixed model, ML becomes REML.

REML

- REML is likelihood approach
- It involves estimation of random effects after accounting for fixed effects and degrees of freedom associated with fixed effects
- This is the major difference REML and maximum likelihood estimates and ensures that REML estimates are not biased.
- The estimates of σ^2_e and σ^2_u are chosen to maximise likelihood and this usually involves <u>iterative schemes</u> involving the first and second differentials.

REML – iterative procedure

(1) use starting value of λ to estimate effect solutions from MME (random – BLUP; fixed – BLUE)

(2) use effect solutions to obtain new estimates of σ^2_e and σ^2_u

$$\sigma_e^2 = \frac{y'y - y'X\hat{b} - y'Z\hat{u}}{N - r(X)}$$

$$\sigma_u^2 = \frac{\hat{u}'A^{-1}\hat{u} + trace(C^{22}A^{-1})\sigma_e^2}{q}$$

(3) go to (1) or stop when $(\lambda_{i+1} - \lambda_i) < \delta$ (*predetermined stopping criterion*)

(4) Consider estimates of σ^2_e and σ^2_u from last iteration as final

REML

- The REML procedure for estimating variance components is available in several packages such as ASREML, Genstat, R, SAS etc.